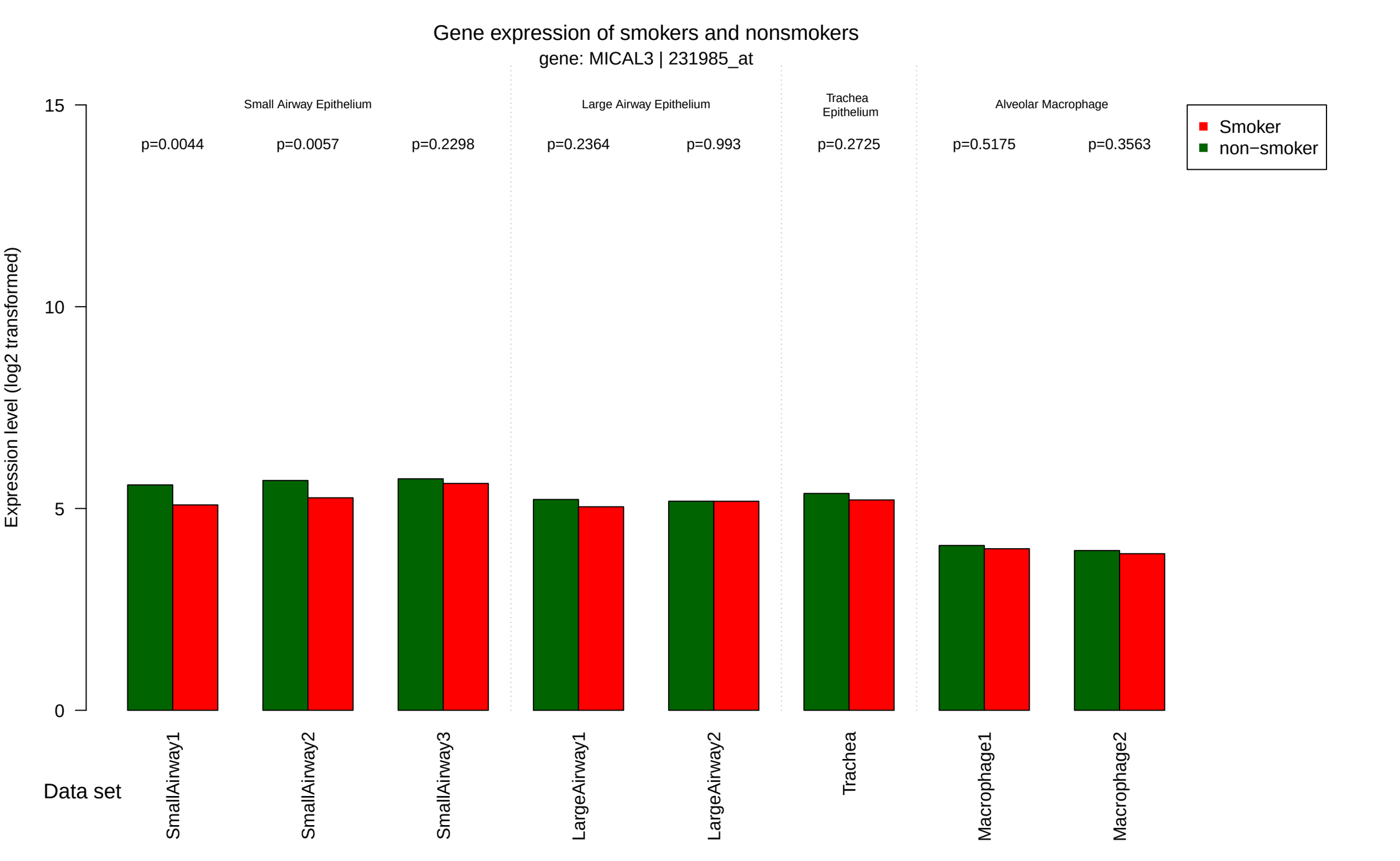
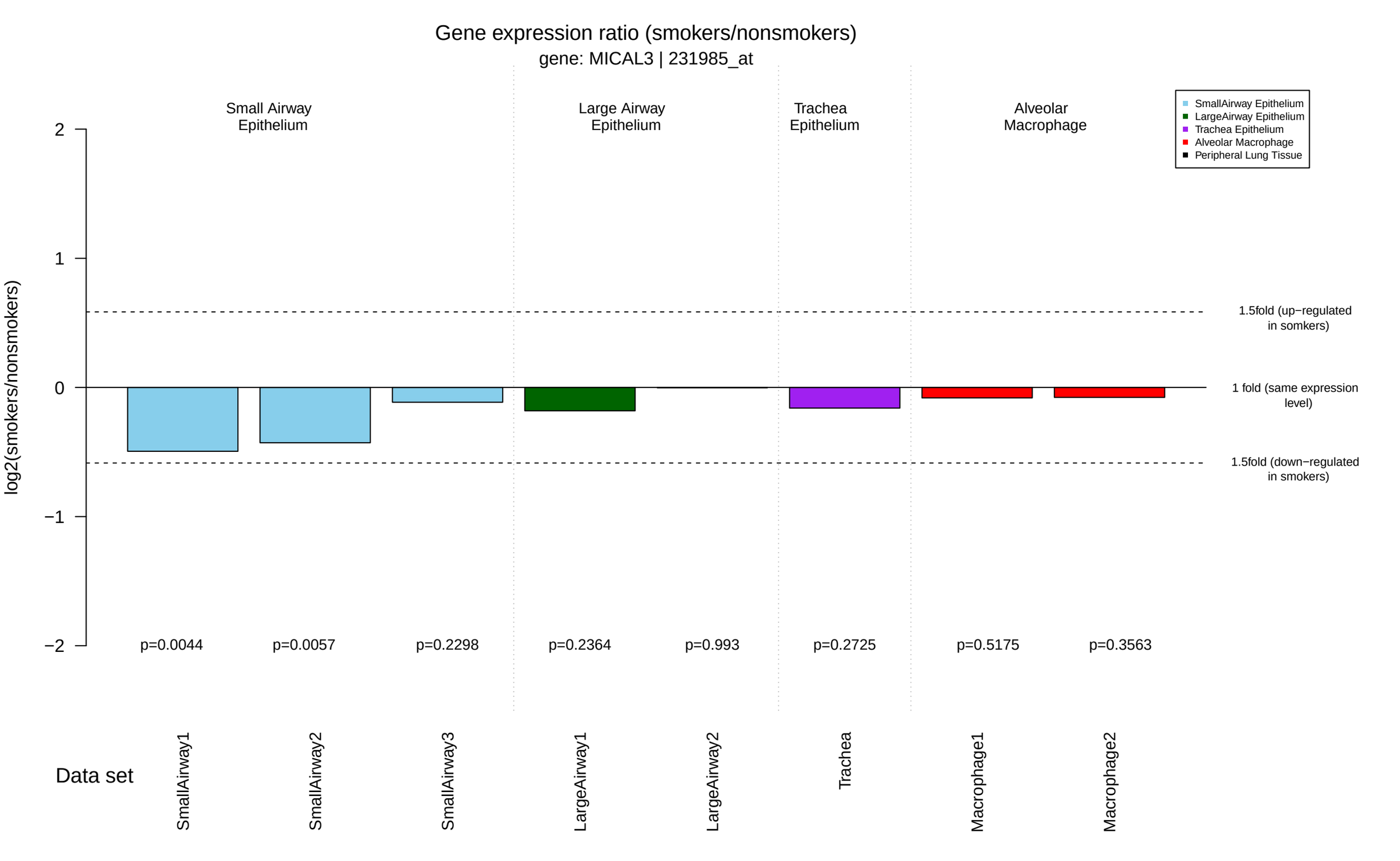
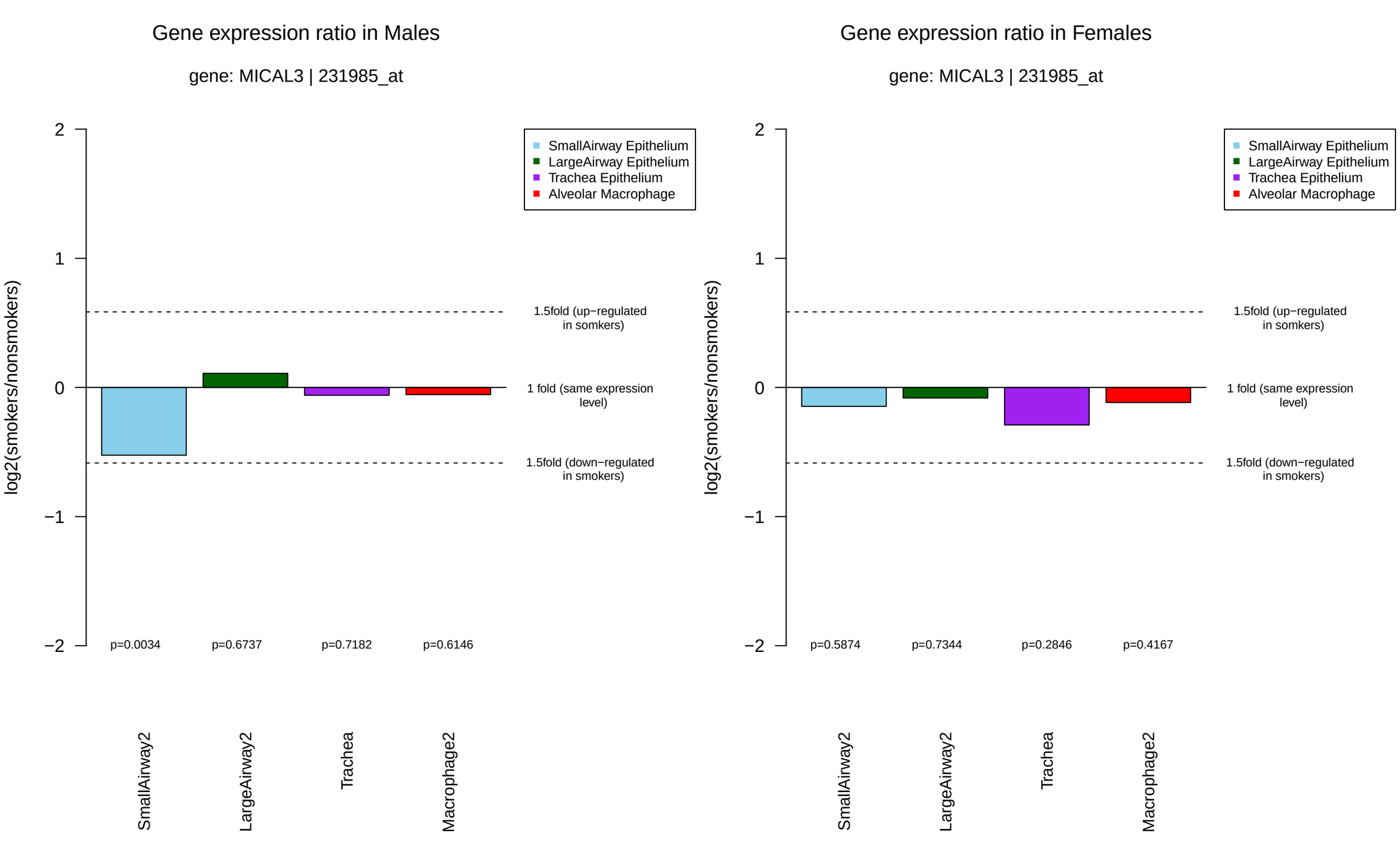
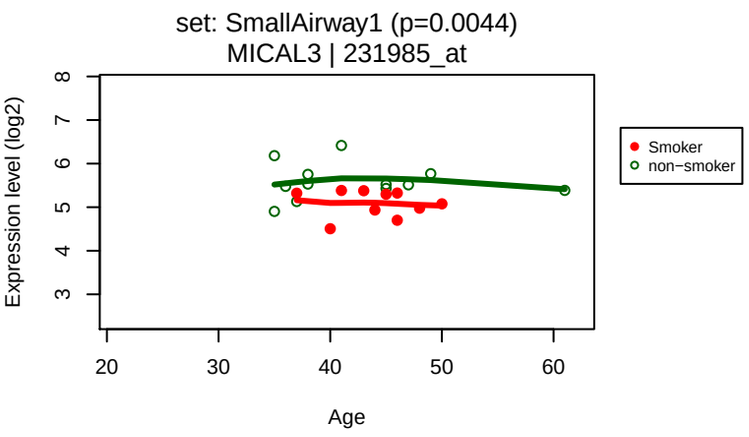
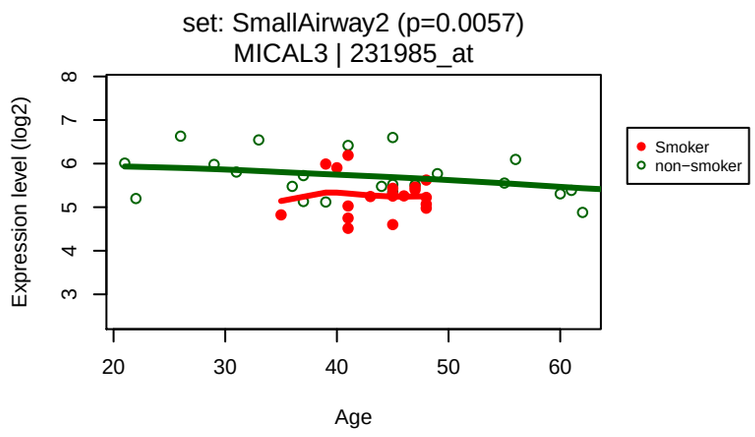
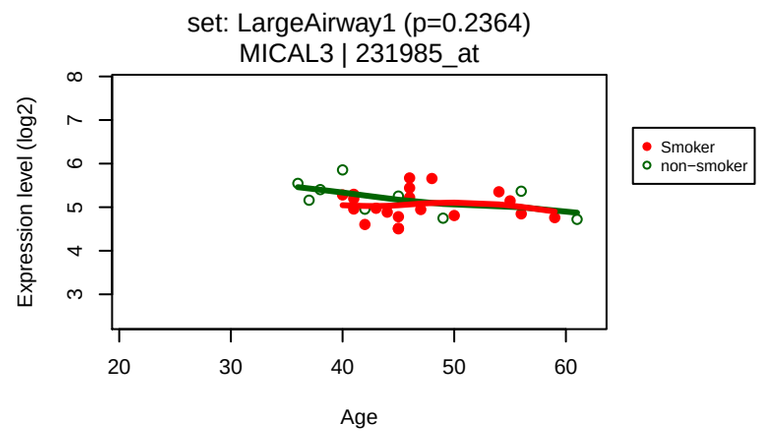

Gene expression of smokers and nonsmokers
gene: MICAL3 | 231985_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0044
p=0.0057
p=0.2298
p=0.2364
p=0.993
p=0.2725
p=0.5175
p=0.3563
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

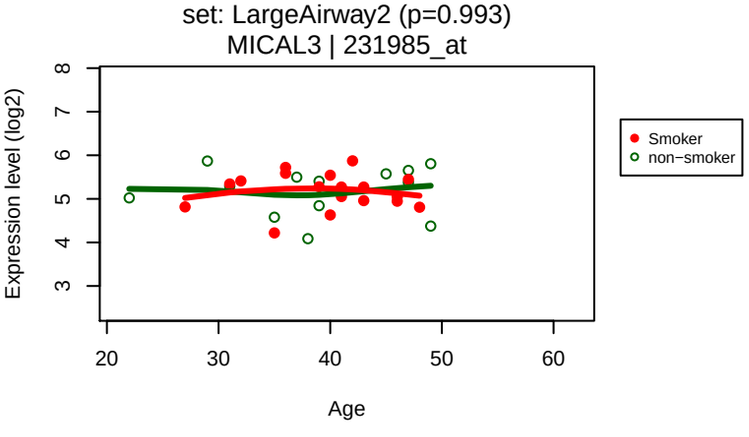

Gene expression ratio (smokers/nonsmokers)
gene: MICAL3 | 231985_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0044
p=0.0057
p=0.2298
p=0.2364
p=0.993
p=0.2725
p=0.5175
p=0.3563
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

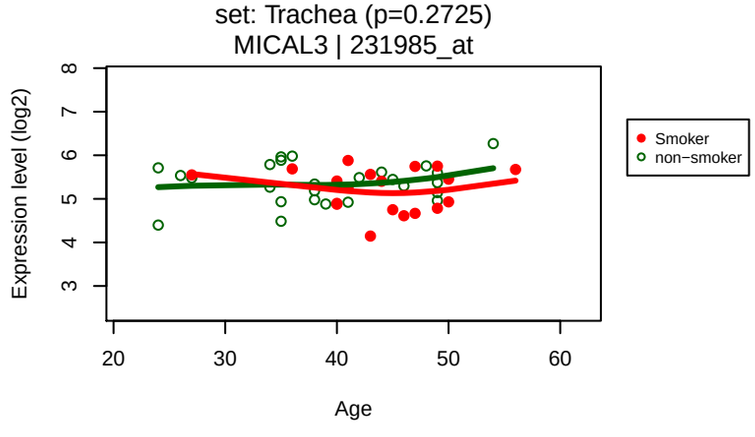

Gene expression ratio in Males
gene: MICAL3 | 231985_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0034
p=0.6737
p=0.7182
p=0.6146
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MICAL3 | 231985_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5874
p=0.7344
p=0.2846
p=0.4167
SmallAirway2
LargeAirway2
Trachea
Macrophage2

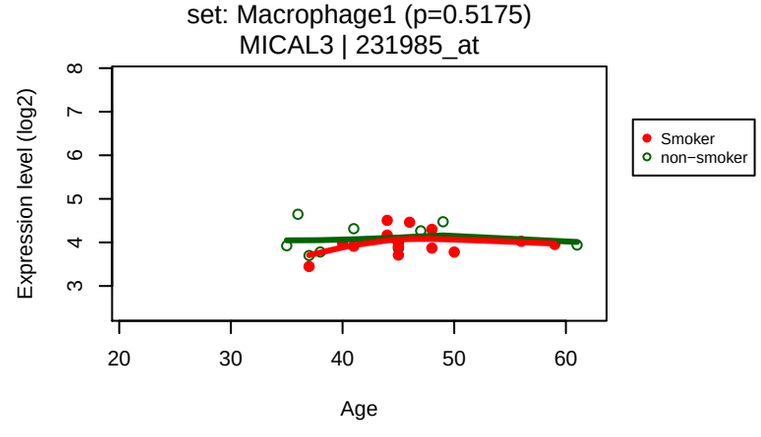

set: SmallAirway1 (p=0.0044)
MICAL3 | 231985_at
Expression level (log2)
Age
Smoker
non−smoker

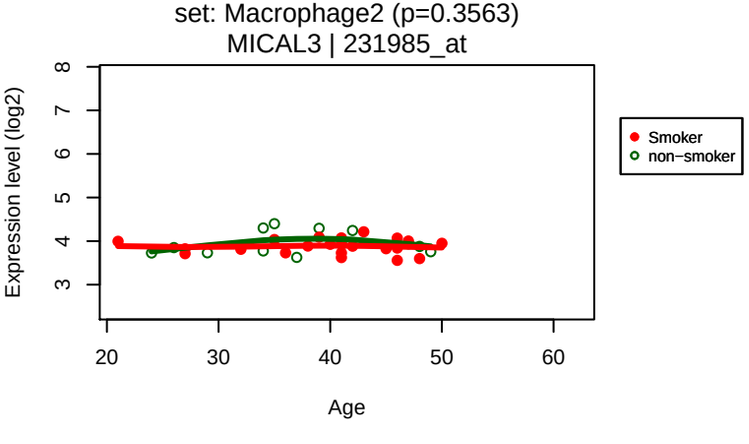

set: SmallAirway2 (p=0.0057)
MICAL3 | 231985_at
Expression level (log2)
Age
Smoker
non−smoker

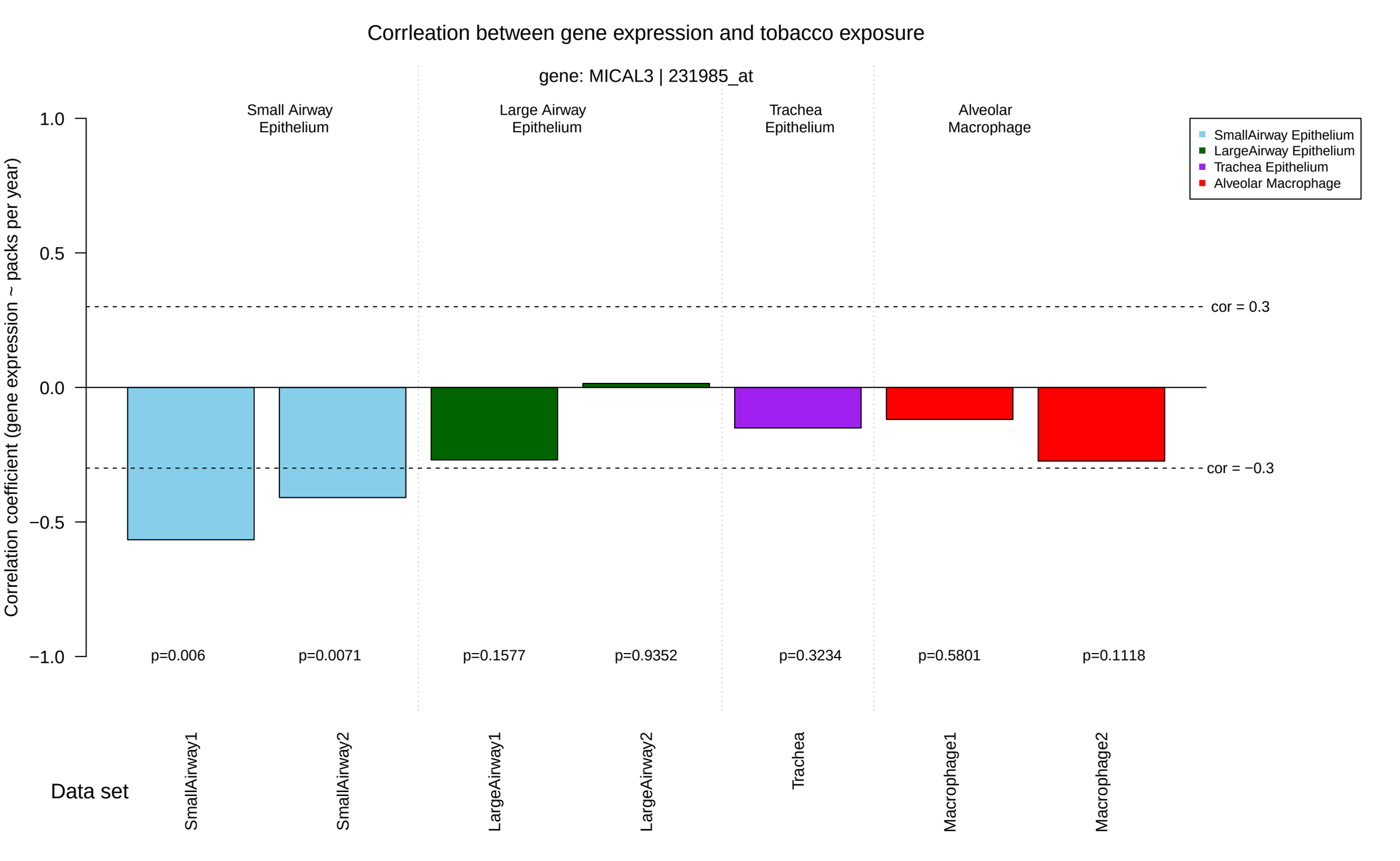

set: LargeAirway1 (p=0.2364)
MICAL3 | 231985_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.993)
MICAL3 | 231985_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2725)
MICAL3 | 231985_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5175)
MICAL3 | 231985_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3563)
MICAL3 | 231985_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MICAL3 | 231985_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.006
p=0.0071
p=0.1577
p=0.9352
p=0.3234
p=0.5801
p=0.1118
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2